

OpenClaw OLED Display Tutorial

Before starting, please complete the peripheral hardware configuration, API Key configuration, model configuration, and three dialogue solution configuration tests in the "Usage Preparation" directory. If you are currently only using TUI / WhatsApp, you can skip the voice interaction configuration. This article assumes the Jetson Nano Docker environment is already prepared and only covers OLED-related content.

1. Tutorial Objectives

This article demonstrates the natural language control capability of OpenClaw on Jetson Nano through the OLED peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach for OpenClaw on Jetson to control OLED display
- Control OLED via WhatsApp, TUI, and voice interaction
- Complete cases including text display, status display, temperature and humidity display, and combined linkage
- Understand the complete chain from "user input" to "screen display"

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw Expansion Board
- [Optional] AI Voice Interaction Module + Speaker

Note:

- OLED screen space is relatively small; it is recommended to display short text.
- Chinese, English, and numbers can all be used for basic text display, but overly long content will be truncated or wrapped.
- If you need to display temperature and humidity, the DHT sensor also needs to be properly connected.

3. Software Preparation

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Jetson-side entry points used in this tutorial:

```
/root/.openclaw/openclaw.json  
/root/.openclaw/workspace/skills/jetson-hardware-control/SKILL.md  
Corresponding atomic tools (servo/rgb/dht/oled/camera/music)  
/root/yahboom_ws/src/largemodel/main.py
```

Common OLED commands include:

```
/usr/local/bin/dht_oled --format json # Display temperature and humidity data on OLED
/usr/local/bin/oled text --text 'Hi,yahboom' #Display text
oled weather --location 'Shenzhen' --condition 'Sunny' --temperature '26°C' --humidity '57%'
#Weather card
/usr/local/bin/oled clear # Clear screen
```

4. Testing OLED Display

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check. After confirming OpenClaw responds normally, proceed with OLED-specific testing.

Enter TUI by typing the following command in the terminal:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔧 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

In the terminal, you can try inputting the following:

- Display "Hi, OpenClaw" on OLED
- Display "Today we start learning OLED" on the screen
- Clear the OLED screen
- Display current status "System OK" on OLED
- Display current temperature and humidity on OLED

```
pi@raspberrypi: ~
File Edit Tabs Help
Display "Hi, OpenClaw" on OLED

"Hi, OpenClaw" is now displayed on the OLED screen. ✓

Display "Today we start learning OLED" on the screen

"Today we start learning OLED" is now displayed on the screen. The text was
automatically wrapped into two lines to fit the OLED display. ✓

Clear the OLED screen

OLED screen cleared. ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
46k/1.0m (5%)
```

5. Three Methods for Dialogue with OpenClaw

Method	Suitable Scenario	Advantages	Recommended Use Case
WhatsApp	Remote control, classroom demonstration	Convenient on mobile, suitable for multi-person experience	Demonstrate remote natural language control
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First establish the control link
Voice Interaction	Voice wake-up, complete AI experience	Closest to real dialogue experience	Final demonstration solution

5.1 WhatsApp Solution

For WhatsApp access, QR code creation, and gateway startup steps, please refer to "Three Dialogue Solution Configuration Tests". This section only includes dialogue examples suitable for the OLED tutorial.

You can directly have conversations, for example:

- Display "Welcome to OpenClaw" on OLED
- Help me display current temperature and humidity data on OLED
- Clear the OLED screen
- Display "System Normal" on the screen

19:52



< 2



+86 153 3885 7526 (你)

给自己发消息

with a nice blue glow.

19:44 ✓✓

Display "welcome to openclaw" on
OLED

19:46 ✓✓

Done! "welcome to openclaw" is now
displayed on the OLED screen. The
text was split across two lines due to
the screen's 128-character width
limitation.

19:47 ✓✓

Display "welcome to openclaw" on
OLED

19:52 ✓✓

Done! "welcome to openclaw" is now
displayed on the OLED screen. The
text was split across two lines due to
the 128-character width limit of the
display.

19:52 ✓✓



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123



space

return



5.2 TUI Solution

If TUI self-check is already complete, directly type in the container:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🚩 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
█
```

It is recommended to first test text display, then test temperature and humidity on screen:

- OLED displays "Hello"
- Clear OLED
- Display current temperature and humidity on the screen

5.3 Voice Interaction Solution

Requires optional AI Voice Interaction Module and speaker.

Execute the following command with openclaw gateway running:

```
cd /root/yahboom_ws/src/largemode1
python3 main.py
```

Suitable for directly demonstrating OLED voice commands, for example:

- Display Hi on OLED
- Change the screen content to "Welcome to OpenClaw"
- Clear the screen

```
2026-05-14 17:25:01 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 8520 ms 语音, 正在转写

[你] Display high on old.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! "high" is now displayed on the OLED screen. 📺
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Clear the screen.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Done! The OLED screen has been cleared. 📺
```

If the voice has recognized the text but OLED has no change, first check whether `openclaw_bridge` has forwarded the voice text to OpenClaw, and whether OpenClaw has generated an OLED action.

6. Comprehensive Cases

The following cases can be executed in TUI, WhatsApp, or voice. It is recommended to use TUI first during the debugging phase.

6.1 Case 1: Text Display

Experiment Objective

Make OLED display a fixed text to complete the most basic screen control.

Effect Description

After OpenClaw receives the command, it will display the specified text on the OLED screen. This case is suitable for verifying if text rendering and screen refresh are normal.

Example Commands

- Display "Hi,yahboom" on OLED
- Display "Today's task is testing OLED"
- Change screen content to "System Ready"

6.2 Case 2: Status Display

Experiment Objective

Make OLED display current status through short text.

Effect Description

This mode is suitable for displaying device status, task status, test results, and other short messages, such as "System OK", "Done", "Ready".

Example Commands

- Display "System OK" on OLED
- Write current status as "Ready"
- OLED displays "Operation Complete"

6.3 Case 3: Temperature and Humidity Display

Experiment Objective

Read the DHT temperature and humidity sensor and display results on OLED.

Effect Description

This case is suitable for environmental monitoring, weather station, or temperature and humidity dashboard basic display. It simultaneously verifies both DHT and OLED peripheral links.

Example Commands

- Display current temperature and humidity on OLED
- Read temperature and humidity and display on screen
- OLED displays current indoor temperature and humidity

6.4 Case 4: Combined Linkage

Experiment Objective

Combine OLED display with other peripheral actions to form a complete feedback process.

Effect Description

OLED can serve as a task result prompt screen. For example, display "Done" after servo action completes, display data after temperature and humidity reading, or display recognition results after visual recognition.

Example Commands

- Turn servo 1 to 90 degrees, finally OLED displays "Complete"
- RGB lights green, then OLED displays "OK"
- Take photo to recognize food, OLED displays food name

7. Common Problems

- OLED has no change: First execute `oled_text --text 'hello'` to confirm if the atomic tool is functioning normally.
- Clear screen ineffective: Execute `oled_clear` for individual testing.
- Temperature and humidity not displaying on screen: First execute `dht` to confirm if DHT reading is normal, then execute `dht_oled` to see if there is normal display.
- Display content too long: Try to use short text, avoid directly displaying long paragraphs on the 128x32 screen.
- Voice recognition normal but no display: Check if `openc1aw_bridge` has forwarded the voice text to OpenClaw.